2025 Year 6 H2 Physics Timed Practice Solutions

1 **C**

$$\text{KE} = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2\left(x_0^2 - \frac{x_0^2}{4}\right) = \frac{1}{2}m\omega^2\left(\frac{3x_0^2}{4}\right)$$

$$\text{PE} = \frac{1}{2}m\omega^2x^2 = \frac{1}{2}m\omega^2\left(\frac{x_0^2}{4}\right)$$

$$\frac{\text{KE}}{\text{PE}} = \frac{\frac{1}{2}m\omega^2\left(\frac{3x_0^2}{4}\right)}{\frac{1}{2}m\omega^2\left(\frac{x_0^2}{4}\right)} = 3$$

2 **C** When the amplitude of vibration of car becomes larger, it implies that it is experiencing resonance.

Natural frequency of suspension system of car = driving frequency on car due to speed bumps.

Using $v = f\lambda$, the separation between bumps is the distance between two peaks of speed bumps i.e., λ.

$$\therefore \lambda = \frac{v}{f} = \frac{5.0}{1.20} = 4.17 \text{ m}$$

3 **B** Y undergoes greater damping than X. Hence, it has a lower peak at resonance and amplitudes at all other driving frequencies are also lower. Additionally, the natural frequency of its oscillation is smaller compared to that of X.

4 **C** 3 × (Period of S) = 30 ms. Hence period of S = 10 ms.

At $t = \frac{1}{4} \times$ period of S = 2.5 ms.

Amplitude of S = displacement of R.

$$y = y_0 \sin \omega t = y_0 \sin\left[\left(\frac{2\pi}{T}\right)t\right]$$

$$= 8\sin\left[\left(\frac{2\pi}{30}\right)2.5\right] = 4 \text{ cm}$$

5 **A** Beyond A, the light intensity is always I.
B allows all of the light from A go through, implies that the axes of polarization of A and B are parallel.
With C between A and B:
angle between the axis of A and the axis of C = angle between the axis of C and the axis of B = θ.
Beyond C, applying Malus' Law: $I_C = I\cos^2\theta$
Beyond B, applying Malus' Law:

$$I_B = I_C \cos^2\theta$$

$$\frac{2I}{3} = \left(I\cos^2\theta\right)\cos^2\theta = I\cos^4\theta$$

$$\cos^4\theta = \frac{2}{3}$$

$$\theta = \cos^{-1}\left(\frac{2}{3}\right)^{\frac{1}{4}} = 25.4°$$

6 **D** Using Rayleigh criterion, $\theta_{min} = \frac{\lambda}{D}$ where D is the diameter the dish of the radio telescope.

Separation between sources $s = r\theta_{min}$.

$$\therefore D = \frac{r\lambda}{s} = \left(\frac{1.0\times10^{15}\times0.030}{1.0\times10^{12}}\right) = 30 \text{ m}$$

7 **C** At the intersection of the path of the microphone and the line joining the 2 loudspeakers, the path difference from the loudspeakers to microphone is 3 times that of the wavelength. Thus those 2 points are the 3rd order maxima of the interference.
Therefore, there must be 12 maxima in total.

8 **C** For diffraction grating: $n\lambda = d\sin\theta$ or $\sin\theta = n\lambda/d$
for $n = 1$, $\sin(15.4°) = \lambda/d$ (1)
for $n = 2$, $\sin\theta_2 = 2\lambda/d$ (2)
(2) ÷ (1): $\sin\theta_2 = 2\sin(15.4°)$
$\theta_2 = 32.8°$
Angle between first and second maxima $= 32.8° - 15.4° = 16.7°$

9 **D** For an ideal gas, $pV = nRT$. Since T is constant, pV is constant for all values of p.

10 **B**

$$W = -\text{enclosed area}$$
$$= -\left[(5-2)10^{-4}\right]\left[(1.5-1.0)10^{5}\right]$$
$$= -15 \text{ J}$$

11 **B** As the change is sudden, there is no time for heat exchange, hence, $q = 0$. As gas expands (volume increases), $w < 0$.

The gas cools, hence, $\Delta U < 0$ (as U is dependent on temperature of gas) **or** since $q = 0$ and $w < 0$, $\Delta U < 0$ $(\because \Delta U = q + w)$.

12 **C**

$$F = \frac{Qq}{4\pi\varepsilon_0 r^2}$$
$$= \frac{\left(1.5\times10^{-17}\right)\left(3.2\times10^{-19}\right)}{4\pi\left(8.85\times10^{-12}\right)\left(1.0\times10^{-13}\right)^2}$$
$$= 4.3 \text{ N}$$

13 B The upward force on the negative charge is of the same magnitude as the downward force on the positive charge. Hence, the net force on the molecule is zero and the velocity remains constant.

14 D Consider vertical forces,

$$F_{\text{net}} = 0$$

$$kv_0 + qE = mg \quad (\text{where } k \text{ is a constant})$$

$$v_0 = \frac{mg}{k} - \left(\frac{q}{k}\right)E$$

Hence, $v_0 - E$ graph is a straight line with negative gradient (and positive vertical intercept).

15 A

$$I = nAvq \quad \text{---(1)}$$

$$2I = n(4A)v_{new}q \quad \text{---(2)}$$

$$\frac{(2)}{(1)}: \quad 2 = \frac{4v_{new}}{v} \quad \Rightarrow \quad v_{new} = \frac{v}{2}$$

16 B

$$P = \frac{V^2}{R}$$

$$R_A = \frac{120^2}{20} = 720\ \Omega$$

$$R_B = \frac{6^2}{20} = 1.8\Omega$$

$$\frac{R_A}{R_B} = \frac{\rho \dfrac{l}{\pi\left(\dfrac{d_A{}^2}{4}\right)}}{\rho \dfrac{l}{\pi\left(\dfrac{d_B{}^2}{4}\right)}} \quad \Rightarrow \quad \frac{R_A}{R_B} = \left(\frac{d_B}{d_A}\right)^2 \quad \Rightarrow \quad \frac{d_A}{d_B} = \sqrt{\frac{R_B}{R_A}} = \sqrt{\frac{1.8}{720}} = 0.050$$

17 B As temperature of thermistor increases, resistance of thermistor decreases. Total resistance in circuit decreases, thus current I_1 increases and hence p.d. across R_1 increases. Therefore, potential difference across thermistor decreases and I_2 decreases.

18 B By potential divider principle,

$$V_{XY} = \frac{8.30}{8.30 + 1.42} \times 9.00 = 7.69\ \text{V}$$

$$E = \frac{0.745}{1.00} \times 7.69 = 5.73\ \text{V}$$

19 A

$$P = I_{rms}{}^2 R = \left(\frac{5}{\sqrt{2}}\right)^2 (10) = 125\ \text{W}$$

20 **D** $V_s = V_p \dfrac{N_s}{N_p} = (20)\left(\dfrac{1000}{200}\right) = 100 \text{ V}$

since transformer is ideal,

$\langle P_p \rangle = \langle P_s \rangle$

$I_p V_p = \dfrac{V_s^2}{R}$

$I_p = \dfrac{1}{20}\left(\dfrac{100^2}{100}\right) = 5.0 \text{ A}$

Peak primary current $= \sqrt{2}(5.0) = 7.1 \text{ A}$

21 **D** As the ring has a gap, there is no induced current . Hence, there is no magnetic field induced and the ring remains stationary.

22 **D** Magnetic force provides centripetal force,

$$B(2q)v = \frac{mv^2}{r}$$

$$B = \frac{mv}{2qr}$$

$$B'qv = \frac{mv^2}{0.5r}$$

$$B' = \frac{2mv}{qr} = 4B$$

23 **A** The induced current in the inner loop should be decreasing in the clockwise direction. Based on Lenz's law, the direction of induced current should be in same direction to oppose the change in B-field from outer loop. The induced current is proportional to the gradient of current vs. time graph.

24 **C**

$$E = \frac{\Delta NBA}{\Delta t}$$

$$= \frac{30\left(60\times10^{-3} - 20\times10^{-3}\right)\left(\pi\times0.035^2\right)}{0.50}$$

$$= 9.2\times10^{-3} \text{ V}$$

$$= 9.2 \text{ mV}$$

25 **A** Energy gap, $\Delta E = hf \;\Rightarrow\; f \propto \Delta E.$

26 **A** Since tube Q has the lower minimum wavelength and higher intensity, it has the higher voltage applied to it. Since the peaks are at the same wavelengths, it is the same material.

27 **C**

$$E = \frac{p^2}{2m}$$

$$1.00\times1.6\times10^{-19}\times10^{6} = \frac{p^2}{2\times1.67\times10^{-27}}$$

$$p = 2.312\times10^{-22} \text{ kg m s}^{-1}$$

$$\Delta p = 0.010\times2.312\times10^{-20} = 2.312\times10^{-22} \text{ kg m s}^{-1}$$

$$\Delta x \Delta p = h$$
$$\Delta x \times 2.312 \times 10^{-22} = 6.63 \times 10^{-34}$$
$$\Delta x = 2.87 \times 10^{-12} \text{ m}$$

28 **D**

29 **A** energy released = (39.25) + (28.48) – (64.94) = 2.79 MeV
γ -ray energy = 2.79 − 2.31 = 0.48 MeV

30 **A** α particles are easily stopped. So β or γ particles are a better choice. To prevent harm to human and the environment, activity should fall quickly.

31 **(a)** $Mg = kx$ or k = gradient $\times g$ M1

$k = \dfrac{0.150 \times 9.81}{0.100} = 14.7 \text{ N m}^{-1}$ M1

(b) **(i)** $\omega = 2\pi f$

$f = \dfrac{1}{2\pi}\sqrt{\dfrac{14.715}{0.450}}$ M1

$= 0.910 \text{ Hz}$ A1

(ii) for an oscillation with amplitude 20.0 cm at displacement of 10.0 cm.

$v = \omega\sqrt{{x_0}^2 - x^2}$

$= 2\pi(0.910)\sqrt{0.200^2 - 0.100^2}$ M1

$= 0.990 \text{ m s}^{-1}$ A1

OR

When length of spring increases from 50.0 cm to 80.0 cm,

decrease in GPE $= Mg \times$ extension $= 0.450 \times 9.81 \times 0.300 = 1.32435$ J

increase in EPE $= \dfrac{1}{2}k({e_2}^2 - {e_1}^2) = \dfrac{1}{2}(14.715)(0.400^2 - 0.100^2) = 1.103625$ J

increase in KE + increase in EPE = decrease in GPE

$$\frac{1}{2}(0.450)v^2 + 1.103625 = 1.32435$$

$$v = 0.990 \text{ m s}^{-1}$$

(c) The mass of the spring-mass system increases and frequency decreases. A1

32 **(a)** $v = f\lambda$

$= 86 \times 4.0$ M1

$= 344 \text{ m s}^{-1}$ A1

(b) 344 m s^{-1} A1

(c) **(i)** The waves from the two loudspeakers travelling in opposite directions undergo superposition to produce the stationary wave since they are of the same frequency, wavelength and amplitude. B1 B1

(ii) inter-nodal distance $= \frac{\lambda}{2}$

$= \frac{4.0}{2}$ M1

$= 2.0$ m A1

(iii) Pressure varies the least at the displacement antinode, B1
since the molecules move such that the distance between neighbouring air molecules is fairly constant over time. B1

33 **(a)** The specific latent heat of vaporisation is the quantity of heat per unit mass required to convert liquid to gas B1
without any change of temperature. B1

(b) **(i)** mass = 424 – 392 = 32 g B1

(ii) temperature difference (between liquid and surroundings) remains the same. B1

(iii) $Q = mL + h$

$P = \left(\frac{m}{t}\right)L + \frac{h}{t}$

From 0 to 10 min, $1.2 \times 230 = \frac{56}{10.0 \times 60}L + \frac{h}{t}$ ---(1) M1

From 12 to 22 min, $1.0 \times 190 = \frac{32}{10.0 \times 60}L + \frac{h}{t}$ ---(2) M1

(1) – (2):

$1.2 \times 230 - 1.0 \times 190 = \frac{56 - 32}{10.0 \times 60}L$ A1

$L = 2150 \text{ J g}^{-1}$

(iv) substitute $L = 2200 \text{ J g}^{-1}$ into (1)

$1.2 \times 230 = \frac{56}{10.0 \times 60} \times 2150 + \frac{h}{t}$

$\frac{h}{t} = 75.3 \text{ W}$ A1

34 **(a)** Electric field strength is numerically equal to potential gradient. A1

(b) for potential to be zero, one potential must be positive and the other potential must be negative / the charges must have opposite signs. M1
electric field strength due to two charges of opposite sign are in the same direction and the electric field strength cannot be zero A1

Alternatively,
for field to be zero, the fields (due to X and Y) must be in opposite directions / the charges must have the same sign. M1
electric potential due to charges of same sign is not zero (with the same sign) A1

Alternatively,
for potential to be zero, one potential must be positive and the other potential must be negative / the charges must have opposite sign. M1
for field to be zero, the fields (due to X and Y) must be in opposite directions / the charges must have the same sign. A1
the signs of the charges cannot (simultaneously) be both the same and opposite (so not possible).

(c) **(i)**

$$V_X + V_Y = 0$$

$$\frac{Q}{4\pi\varepsilon_0 x} + \frac{-2Q}{4\pi\varepsilon_0 y} = 0$$ M1

$$y = 2x$$ A1

(ii)

$$E_X = \frac{Q}{4\pi\varepsilon_0 x^2}$$ A1

(iii)

$$\begin{aligned} E &= E_X + E_Y \\ &= \frac{Q}{4\pi\varepsilon_0 x^2} + \frac{2Q}{4\pi\varepsilon_0 y^2} \\ &= \frac{Q}{4\pi\varepsilon_0 x^2} + \frac{2Q}{4\pi\varepsilon_0 (2x)^2} \\ &= \frac{3Q}{8\pi\varepsilon_0 x^2} \end{aligned}$$ A1

35 **(a)** It is the amount of electrical energy per unit charge that is converted from other forms of energy. B1

(b) **(i)** When the switch is open, there is no current in the circuit and the reading is the e.m.f. of the cell. B1
When the switch is closed, there is current in the circuit and there is a potential drop/difference across the internal resistance of the cell. B1
The reading is now the difference between the e.m.f. and potential difference.

(ii) Effective resistance of 40 Ω and 60 Ω resistors:

$$\frac{1}{R_{||}} = \frac{1}{40} + \frac{1}{60}$$ B1

$$R_{||} = 24\ \Omega$$

$$V = IR$$

$$11.4 = I(14 + 24)$$ M1

$$I = 0.30\ \text{A}$$ A1

(iii)

$$V = E - Ir$$

$$11.4 = 12.0 - 0.30r$$ M1

$$r = 2.0\ \Omega$$ A1

36 **(a)** $E = \left| -\dfrac{\Delta V}{\Delta d} \right|$

$= \dfrac{3000-(-600)}{8.0\times10^{-3}}$ M1

$= 4.5\times10^{5}\ \text{V m}^{-1}$ A1

(b) $F_B = F_E$

$Bev = eE$

$B = \dfrac{E}{v}$ M1

$= \dfrac{4.5\times10^{5}}{7.0\times10^{6}}$

$= 0.0643\ \text{T}$ A1

(c) **(i)** Magnetic force provides centripetal force

$$Bqv = \frac{mv^2}{r}$$

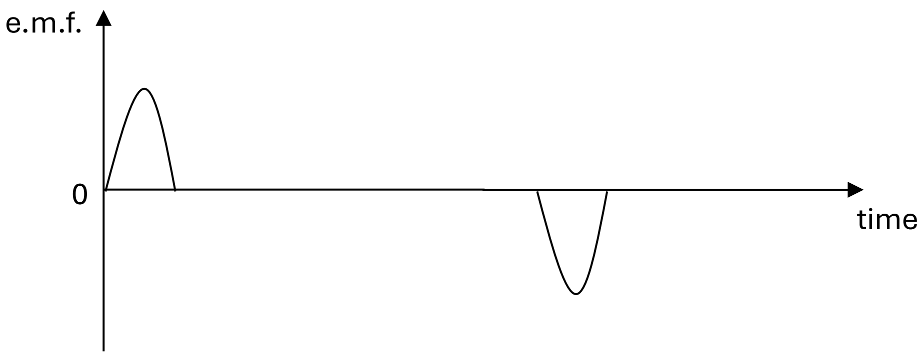
$$\left(3.5\times10^{-3}\right)\left(1.6\times10^{-19}\right) = \frac{\left(9.11\times10^{-31}\right)\left(7.0\times10^{6}\right)}{r}$$ M1

$$r = 0.0114\ \text{m}$$ A1

(ii) The magnetic force acts perpendicularly to the velocity of the electrons (i.e. no acceleration in the direction of the velocity). M1
Hence, its speed remains the same. A1

37 **(a)** Faraday's law of electromagnetic induction states that the induced e.m.f. is proportional to the rate of change of magnetic flux linkage. B1

(b) **(i)**



B1

Peaks in opposite direction and zero between peaks
Peaks have same area.

(ii) As the coil first enters the field, the magnetic flux linkage increases at an increasing rate until half of the coil is in the field. By Faraday's law, e.m.f. increases. B1
After half the coil is in the field, magnetic flux linkage increases at a decreasing rate. Hence e.m.f. decreases. B1
When the entire coil is inside the magnetic field, there is no change of magnetic flux linkage. Hence, the induced e.m.f. is zero. B1

(iii) 1. maximum e.m.f. for both peaks increases B1
2. time interval between peaks decreases / peaks are thinner (because area under each peak (i.e. $\Delta\Phi$) is constant). B1

(iv) At 2.0 s, half of the coil had entered the magnetic field.

$$\langle E \rangle = \left| -\frac{d(N\phi)}{dt} \right|$$

$$= \frac{NBA - 0}{t}$$ M1

$$= \frac{(500)(3.0\times10^{-4})(0.50\times\pi\times0.040^2)}{2.0}$$ A1

$$= 1.88\times10^{-4}\text{ V}$$

38 (a) Photoelectric effect is the <u>emission of electrons from the surface</u> of a metal when it is exposed to <u>electromagnetic radiation of sufficiently high frequency</u>. B1

(b) (i)

$$f_{threshold} = \frac{\phi}{h}$$

$$= \frac{2.0\times(1.6\times10^{-19})}{6.63\times10^{-34}}$$

$$= 4.83\times10^{14}\text{ Hz}$$ B1

(ii)

$$hf = eV_s + \phi$$

$$V_s = \frac{hf - \phi}{e}$$

$$= \frac{(6.63\times10^{-34})(\frac{3\times10^8}{420\times10^{-9}}) - \left[2.0\times(1.6\times10^{-19})\right]}{1.60\times10^{-19}}$$ M1

$$= 0.960\text{ V}$$ A1

(iii) $I = n_e e$,where n_e is the number of photoelectrons per second

$$n_e = \frac{I}{e} = \frac{4.8\times10^{-10}}{1.60\times10^{-19}} = 3.0\times10^9\text{ s}^{-1}$$ B1

(iv) $n_p = 2500 n_e$,

where n_p is the number of photons per second.

$$I = \frac{P}{A}$$

$$= \frac{n_p hf}{A} = \frac{n_p hc}{A\lambda}$$

$$= \frac{(2500)(3.0\times10^9)(6.63\times10^{-34})(3\times10^8)}{(25\times10^{-6})(420\times10^{-9})}$$

$$= 0.142\text{ W m}^{-2}$$ A1

M1 for $I = \frac{P}{A}$ and correct substitution

M1 for $P = n_p hf$ and correct substitution

39 **(a)** Half-life is the time taken for the number of undecayed nuclei to be reduced to half its original number. B1

(b)

$$\left.\begin{aligned}\lambda &= \frac{\ln 2}{t_{\frac{1}{2}}} \\ &= \frac{\ln 2}{5730} \\ &= 1.2097 \times 10^{-4}\ \text{y}^{-1} \\ &= 3.8359 \times 10^{-12}\ \text{s}^{-1}\end{aligned}\right\}$$ M1

Consider 1 g of carbon-14,

$$A = \lambda N$$

$$0.255 = (3.8359 \times 10^{-12})\,N$$

$$N = 6.65 \times 10^{10}$$ A1

(c) For 500 mg of carbon,

$$A_0 = M\left(\frac{A}{m}\right) = (0.500)\,(0.255)\,3600 = 459\ \text{hour}^{-1}$$ M1

$$A = A_0\, e^{-\lambda t}$$

$$174 = (459)\, e^{-1.2097 \times 10^{-4}\, t}$$ M1

$$t = 8020\ \text{years}$$ A1